21-242 Recitation 12

Rank–nullity — Q3. T -1(U) U+ T

Course-level solution, with a later-course alternative when useful

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Prompt

Let $T : V \to W$ and $U \subseteq W$ a subspace. Show that $\dim(T^{-1}(U)) \leq \dim(U) + \dim(\ker(T))$. Find examples of when equality does and doesn't hold.

Course-level solution

Assume $\dim V < \infty$ (so all dimensions below are finite). Restrict the codomain: the restriction

$$T\big|_{T^{-1}(U)} : T^{-1}(U) \longrightarrow U \cap \operatorname{im}(T)$$

is linear, surjective (by definition of image), and has kernel $\ker T \cap T^{-1}(U) = \ker T$. Rank-nullity therefore gives

$$\dim T^{-1}(U) = \dim\big(U \cap \operatorname{im}(T)\big) + \dim \ker T \leq \dim U + \dim \ker T,$$

using $U \cap \operatorname{im}(T) \subseteq U$.
Equality holds if and only if $U \cap \operatorname{im}(T) = U$, i.e. iff $U \subseteq \operatorname{im}(T)$.
Equality example. Let $T = \mathrm{id}_V : V \to V$ and let $U \subseteq V$ be any subspace. Then $\ker T = \{\vec{0}\}$ and $T^{-1}(U) = U$, so $\dim T^{-1}(U) = \dim U = \dim U + \dim \ker T$.
Strict inequality example. Let $T = 0 : V \to W$ with $\dim V \geq 1$ and let $U \subseteq W$ satisfy $\dim U \geq 1$ (e.g. $U = W$ if $\dim W \geq 1$). Then $T^{-1}(U) = V$, and

$$\dim V \leq \dim U + \dim V$$

is equality only if $\dim U = 0$. For $\dim U > 0$ we have a strict inequality: $\dim T^{-1}(U) = \dim V < \dim U + \dim V$.
A mixed example: inclusion $T : \mathbb{F}^2 \to \mathbb{F}^3$, $T(x, y) = (x, y, 0)$, and $U = \mathrm{Span}\{e_3\}$. Then $U \cap \operatorname{im} T = \{\vec{0}\}$, $T^{-1}(U) = \ker T = \{\vec{0}\}$, and $0 < 1 + 0$.

Later-course solution (using ideas from the rest of the handout)

The identity $\dim T^{-1}(U) = \dim(U \cap \operatorname{im} T) + \dim \ker T$ is the precise sharpening; the inequality in the prompt is what you get by crudely estimating $\dim(U \cap \operatorname{im} T) \leq \dim U$.